OVIYA ELANGOVAN

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EDUCATION

Virginia Tech — M.S. Computer Engineering **Aug. 2026 – May 2028**
Blacksburg, VA

Sri Sairam Engineering College — B.E. Electronics and Communication Engineering **2018 – 2022**
Chennai, India | GPA: 3.16/4.00

RESEARCH EXPERIENCE

Virginia Tech — Wage-Supported Graduate Research Assistant **Sep. 2026 – Present**
Pig Body Condition Score Estimation | Blacksburg, VA

- Develop an end-to-end computer vision pipeline using YOLOv8 detection, SAM2 segmentation, and ResNet-18 regression to estimate pig body condition scores from RGB and depth imagery.
- Compare image-resizing and padding strategies using animal-disjoint evaluation; run reproducible GPU experiments on Virginia Tech ARC and analyze MAE, RMSE, R^2, exact accuracy, and within-one accuracy.

Virginia Tech — Volunteer Researcher **Feb. 2026 – Aug. 2026**

- Labeled and quality-checked livestock imagery, developed YOLO-based crop generation workflows, and supported dataset preparation for downstream BCS modeling.

PROFESSIONAL EXPERIENCE

Wright Brothers Analytics — Data Science Intern **Mar. 2024 – Apr. 2025**
Chennai, India

- Cleaned and transformed structured and text datasets, performed exploratory analysis and feature engineering, and built reproducible Python workflows for model development and validation.
- Collaborated on analytics deliverables, automated data-quality checks and ETL tasks, and communicated results through Power BI dashboards and technical documentation.

PROJECTS

Automated Support-Ticket Classification and Routing **Python | NLP | FastAPI**

- Prepared ticket text and labels, trained a support-queue classifier, exposed confidence-scored predictions through FastAPI, and routed uncertain cases for manual review.

Sales and Revenue Performance Dashboard **SQL | Power BI | DAX**

- Modeled sales, customer, product, region, and target data; created Power Query transformations, DAX KPIs, filters, and drill-down visuals for stakeholder reporting.

Cloud Customer Review Sentiment Monitor **AWS | Python | REST API**

- Built a serverless sentiment workflow with API Gateway, Lambda, S3, DynamoDB, and CloudWatch to process reviews, store predictions, and monitor execution health.

INDEPENDENT PROJECTS

Fraud Risk ML Platform GitHub **Python | ML | API**

- Developed an end-to-end fraud-risk scoring platform with preprocessing, model inference, API-based predictions, and structured outputs for reviewing high-risk transactions.

ResolveAI GitHub **Python | RAG | LLM**

- Built a retrieval-augmented question-answering application that retrieves relevant document context and uses a language model to generate grounded responses.

TECHNICAL SKILLS

Languages: Python, SQL, C | **ML and Computer Vision:** Scikit-learn, Random Forest, ResNet-18, YOLOv8, SAM2, OpenCV | **Data and Cloud:** Pandas, NumPy, Power BI, DAX, FastAPI, PostgreSQL, AWS Lambda, S3, DynamoDB, CloudWatch | **Tools:** Git, VS Code, Jupyter Notebook, Google Colab, Slurm